

Homework 04: Data frames and data wrangling

Please read the entire chapter on data transformation from [R for Data Science](#) before starting this homework.

This homework relies on the `nycflights13` package, which contains several data frames, including:

- `airlines`
- `airports`
- `flights`
- `planes`
- `weather`

Loading `nycflights13` puts these data frames on the search path.

Setup

Load packages (do this every time)

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(nycflights13)
```

Note - if you don't have these installed you will need to first `install.packages('nycflights13')`

Question 1: Filtering

Make a plot of **air time vs distance** (air time on the y-axis, distance on the x-axis) for all flights that meet the following criteria:

- originate from LaGuardia airport ("LGA")
- departed on the 16th of the month
- have a flight distance of less than 2000 miles

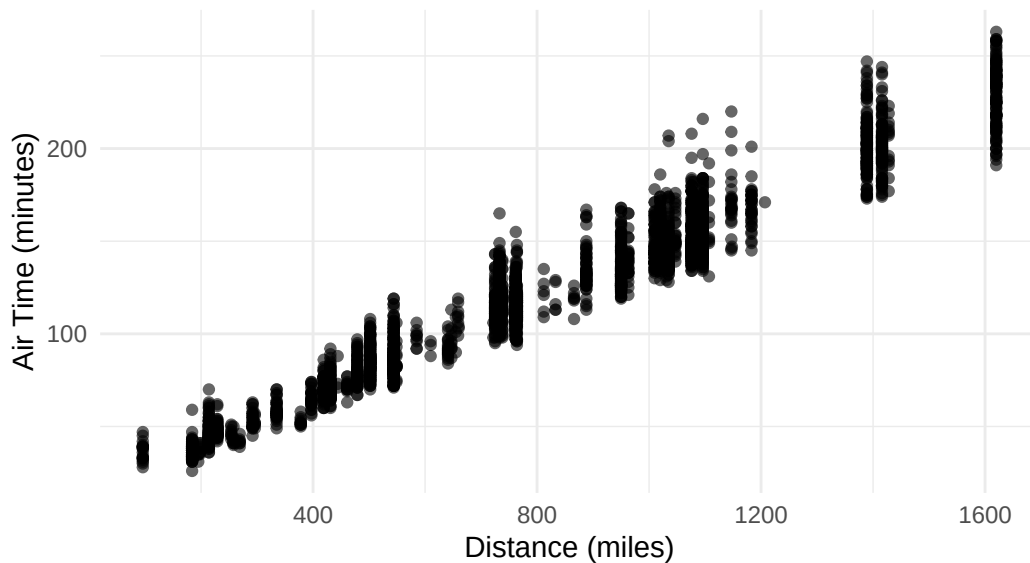
Your code

```
filtered_flights <- flights %>%
  filter(
    origin == "LGA",
    day == 16,
    distance < 2000
  ) %>%
  drop_na(air_time, distance)

ggplot(filtered_flights, aes(x = distance, y = air_time)) +
  geom_point(alpha = 0.6) +
  labs(
    title = "Air Time vs Distance",
    subtitle = "The Graph",
    x = "Distance (miles)",
    y = "Air Time (minutes)"
  ) +
  theme_minimal()
```

Air Time vs Distance

The Graph



Brief written response

In 2–4 sentences, describe what you observe in the plot. (For example: Is the relationship roughly linear? Are there any clear outliers?)

Your answer here: The graph shows a clear positive relationship between distance and air time. As distance increases, air time also increases in a mostly straight-line pattern. There are only a few small outliers, but overall the pattern is very consistent.

Question 2: Dealing with NAs

Make a data frame of all rows of `flights` that have values for *both* `arr_time` and `dep_time` (i.e., neither value is NA).

Your code

```
library(tidyverse)
library(nycflights13)

flights_complete <- flights %>%
  filter(!is.na(arr_time), !is.na(dep_time))

head(flights_complete)
```

```
# A tibble: 6 x 19
   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
  <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
1  2013     1     1     517             515           2     830             819
2  2013     1     1     533             529           4     850             830
3  2013     1     1     542             540           2     923             850
4  2013     1     1     544             545          -1    1004            1022
5  2013     1     1     554             600          -6     812             837
6  2013     1     1     554             558          -4     740             728
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>
```

Filtering NAs (conceptual)

`ggplot()` will automatically remove NA values from a plot, but it emits a warning message about it. You *could* silence warnings using chunk options, but instead:

Brief written response

Explain (in words) how you could prevent those NA values from appearing in the plot in the first place.

Your answer here: To prevent NA values from appearing in the plot, I would filter them out before making the graph. I can use **`filter()`** together with **`!is.na()`** to remove rows where important variables have missing values. This way, `ggplot` only receives clean data and won't show a warning message.

Question 3: Adding columns

Create a data frame of **average flight speeds**, based on `air_time` and `distance`.

Then make either:

- a histogram, **or**
- a density plot

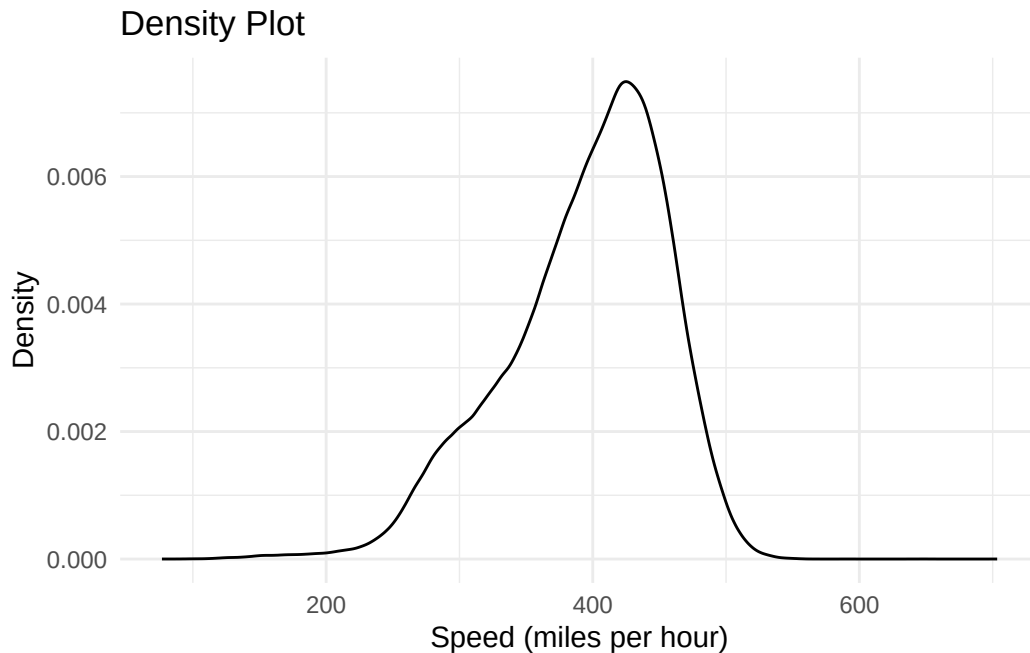
If you like, you may break the data out (e.g., by airline or another variable) in a way that you think makes sense.

Your code

```
library(tidyverse)
library(nycflights13)

speed_df <- flights %>%
  filter(!is.na(air_time), !is.na(distance)) %>%
  mutate(speed = distance / (air_time / 60))

ggplot(speed_df, aes(x = speed)) +
  geom_density() +
  labs(
    title = "Density Plot",
    x = "Speed (miles per hour)",
    y = "Density"
  ) +
  theme_minimal()
```



Brief written response

Describe the main features of your speed distribution.

Your answer here: The density plot shows that most flights have an average speed around 400–450 miles per hour. The curve is slightly skewed to the right, which means a few flights are much faster than most. The speeds are fairly consistent for the majority of flights, with only a small number of unusually slow or fast flights.

Rendering and submission (GitHub)

Canvas contains the course GitHub link and instructions for **forking** the course repository.

In this file, your job is to:

1. Complete the homework by adding code + written responses above.
2. Render this Quarto file to **HTML or PDF**.
3. Move the rendered file(s) to the /doc folder (note that, if you render to html you will also have to move your hmk_04_data_frames_files folder)
4. Add/commit both:

- this edited `.qmd` file
 - the rendered output file(s) (HTML or PDF)
5. Push your changes to your fork on GitHub.
 6. Open a Pull Request back to the course repository.

What to submit on Canvas

On Canvas, submit a link to your GitHub fork repository (not files).